# AWS VPC PEERING LAB

EWB Cloud Labs — Evidence and Completion Report

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| **Batch** | 2027 Batch |
| **Project Title** | AWS VPC Peering with Bastion Access and Private-Subnet Connectivity |
| **AWS Region** | Asia Pacific (Sydney) — ap-southeast-2 |
| **Completion / Report Date** | 29 October 2026 |

This report documents the AWS VPC peering lab resources, routing design, access controls, connectivity tests, and the supplied evidence screenshots. Resource identifiers and addresses below reflect the lab details recorded during execution; screenshots should be treated as the primary visual evidence.

# 1. Objective and Result

Objective: establish private network connectivity between two VPCs using VPC peering, keep application instances without public IPv4 addresses, and use a Bastion host as the SSH entry point.

**Recorded outcome:** the peering connection was Active; SSH access to the Bastion and private instances was tested; ICMP connectivity was tested in both directions; a controlled failure was produced by temporarily removing an inbound ICMP rule; and connectivity recovered after the rule was restored.

# 2. Architecture and Resource Inventory

| Component | Name / ID | CIDR / Address / Notes |
|---|---|---|
| VPC A | ewb-vpc-a | 10.10.0.0/16 |
| Public subnet (VPC A) | ewb-a-public | 10.10.1.0/24 |
| Private subnet (VPC A) | ewb-a-private | 10.10.2.0/24 |
| VPC B | ewb-vpc-b | 10.20.0.0/16 |
| Private subnet (VPC B) | ewb-b-private | 10.20.1.0/24 |
| Internet Gateway | ewb-igw-a / igw-08e9646d38cf32aaf | Attached to VPC A |
| VPC peering | ewb-vpc-a-to-b / pcx-0d38f02965dce57fe | Recorded status: Active |
| Bastion | ewb-bastion / i-09626a7b8d57085c0 | Private: 10.10.1.10 \| Public: 54.253.218.4 (recorded) |
| App-A | ewb-app-a / i-0cf98e4dd39856b91 | Private: 10.10.2.10 \| No public IPv4 |
| App-B | ewb-app-b / i-0c11f219fc5beebed | Private: 10.20.1.10 \| No public IPv4 |
| SSH key pair | ewb-lab | Used for SSH access |

**Logical path:** Mac workstation → SSH to Bastion (public subnet in VPC A) → SSH to App-A or App-B. VPC A and VPC B communicate over the active peering connection using private IP addresses. App-A and App-B do not have public IPv4 addresses.

# 3. Route Tables and Security Controls

| Route table | Destination | Target |
|---|---|---|
| ewb-rt-a-public | 10.10.0.0/16 | local |
| ewb-rt-a-public | 0.0.0.0/0 | igw-08e9646d38cf32aaf |
| ewb-rt-a-public | 10.20.0.0/16 | pcx-0d38f02965dce57fe |
| ewb-rt-a-private | 10.10.0.0/16 | local |
| ewb-rt-a-private | 10.20.0.0/16 | pcx-0d38f02965dce57fe |
| ewb-rt-b-private | 10.20.0.0/16 | local |
| ewb-rt-b-private | 10.10.0.0/16 | pcx-0d38f02965dce57fe |

## Recorded inbound security-group intent

| Security group | Inbound rule | Source |
|---|---|---|
| ewb-sg-bastion | SSH / TCP 22 | Current workstation public IP /32 (verify current value in AWS) |
| ewb-sg-app-a | SSH / TCP 22 | 10.10.1.10/32 |
| ewb-sg-app-a | ICMPv4 Echo Request | 10.20.1.10/32 |
| ewb-sg-app-b | SSH / TCP 22 | 10.10.1.10/32 |
| ewb-sg-app-b | ICMPv4 Echo Request | 10.10.2.10/32 |

Security-group sources and route targets above are recorded from the lab setup. Confirm the live AWS console before cleanup, especially the Bastion source IP and restored App-B ICMP rule.

# 4. Validation and Test Results

| Test | Recorded result |
|---|---|
| SSH to Bastion | Successful |
| SSH to App-A via Bastion | Successful |
| SSH to App-B via Bastion | Successful |
| App-A → App-B ICMP | Successful: 4 sent, 4 received, 0% loss |
| App-B → App-A ICMP | Successful: 4 sent, 4 received, 0% loss |
| Controlled failure | App-A → App-B ping failed while App-B inbound ICMP rule was temporarily removed: 4 sent, 0 received, 100% |
| Recovery | App-A → App-B ping succeeded after restoring the ICMP rule: 4 sent, 4 received, 0% loss |

## Evidence limitation

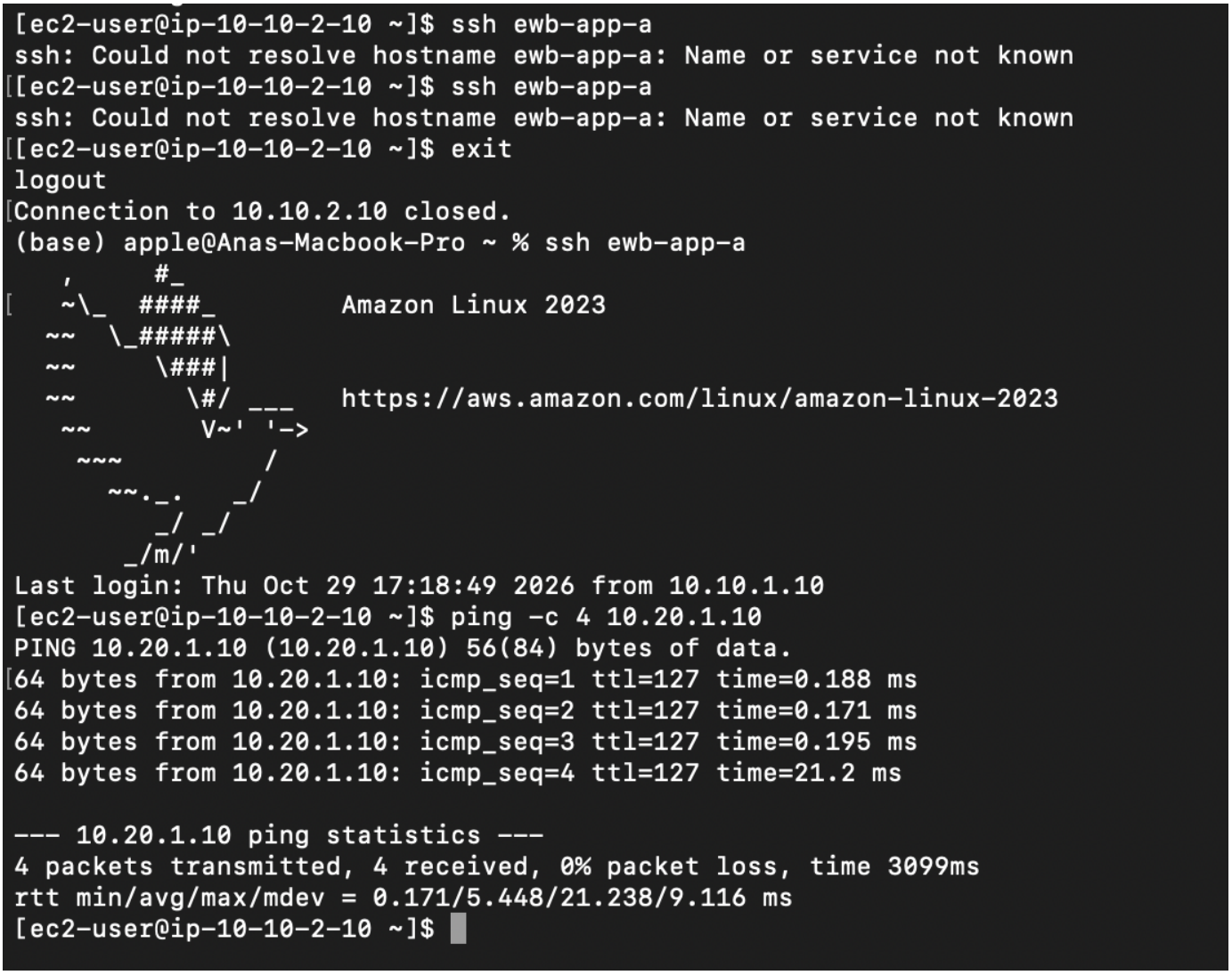
A baseline connectivity screenshot taken before creating the peering connection was not captured. The failure screenshot in this report documents a deliberate security-group ICMP-rule removal after the peering setup, and is labelled accordingly rather than being presented as a pre-peering baseline.

# 5. Screenshot Evidence Appendix

The following screenshots were supplied in screenshots.zip and are included without altering their contents.

## 1. Connectivity test: App-A to App-B

Evidence of the successful ICMP test from App-A (10.10.2.10) to App-B (10.20.1.10).




File: 01-App-A-to-App-B-Success.png

## 2. Connectivity test: App-B to App-A

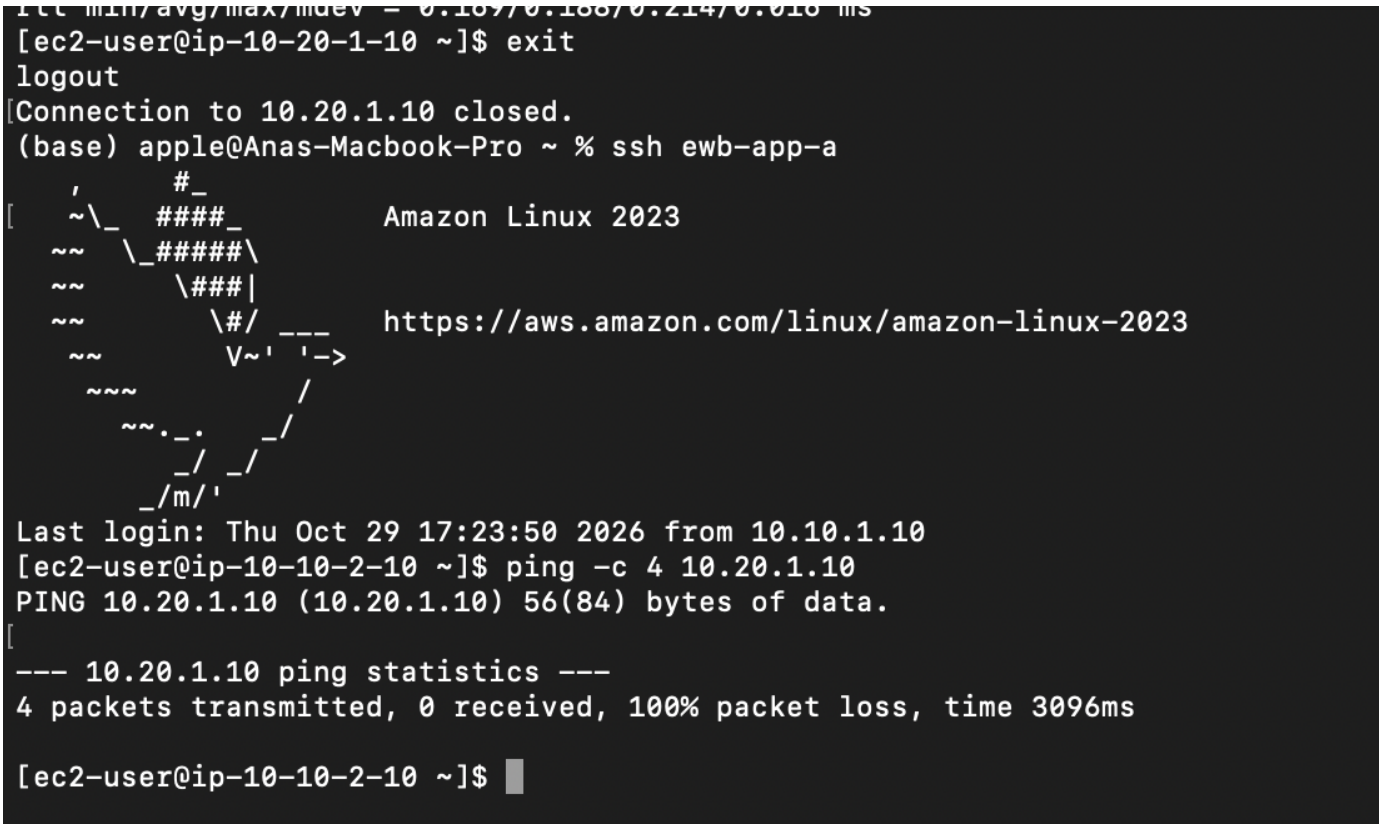
Evidence of the successful ICMP test from App-B (10.20.1.10) to App-A (10.10.2.10).




File: 02-App-B-to-App-A-Success.png

## 3. Controlled failure test

The ping test failed while the relevant inbound ICMP Echo Request rule was temporarily removed. This is a security-group controlled-failure test, not a baseline-before-peering test.




File: 03-Controlled-Failure.png

## 4. Recovery after restoring the rule

The ping test succeeded after the ICMP rule was restored.

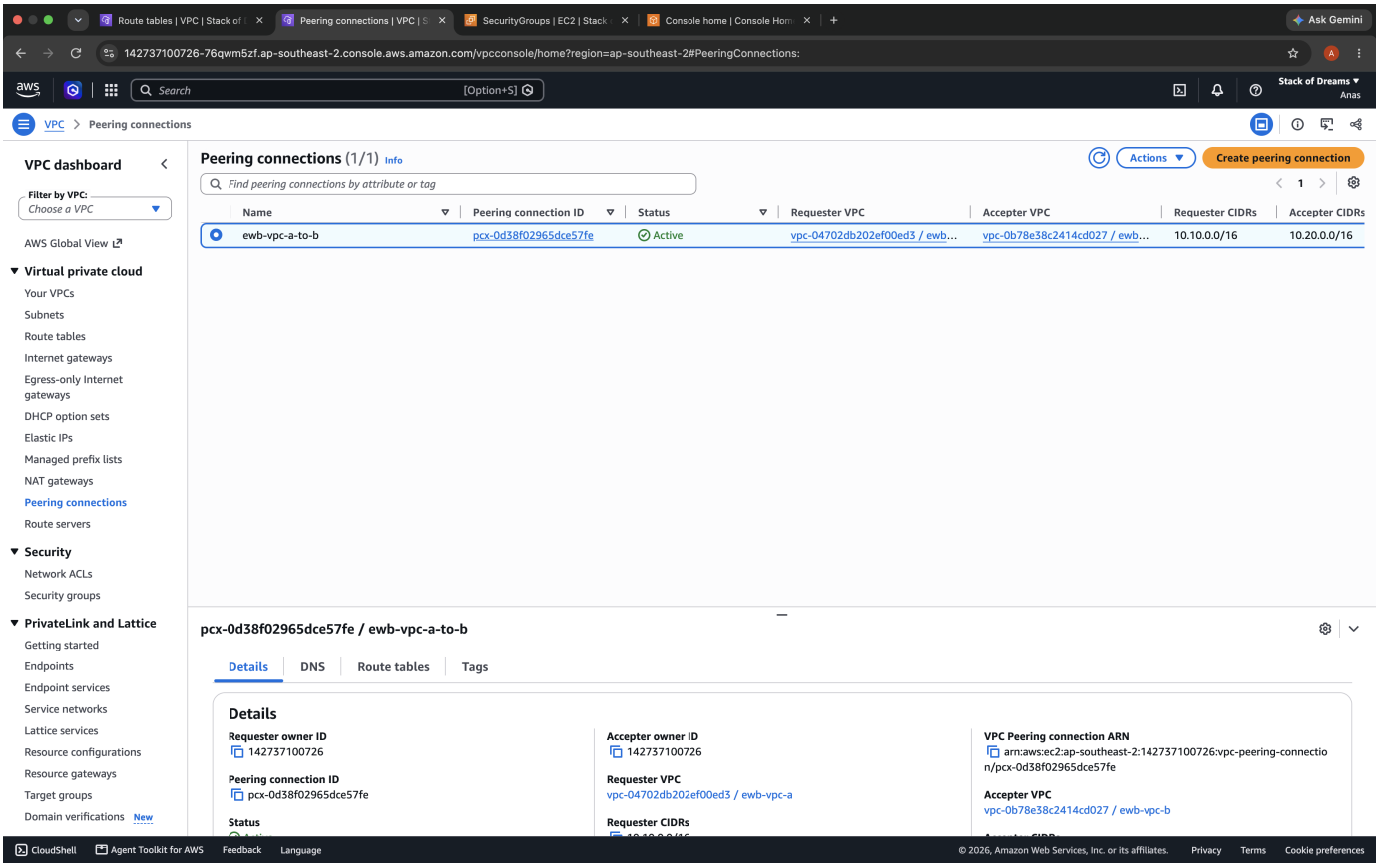
```
(base) apple@Anas-Macbook-Pro ~ % ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Oct 29 17:30:17 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.175 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.179 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.202 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.179 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3096ms
rtt min/avg/max/mdev = 0.175/0.183/0.202/0.010 ms
[ec2-user@ip-10-10-2-10 ~]$
```

File: 04-Recovery-Success.png

# 5. Active VPC peering connection

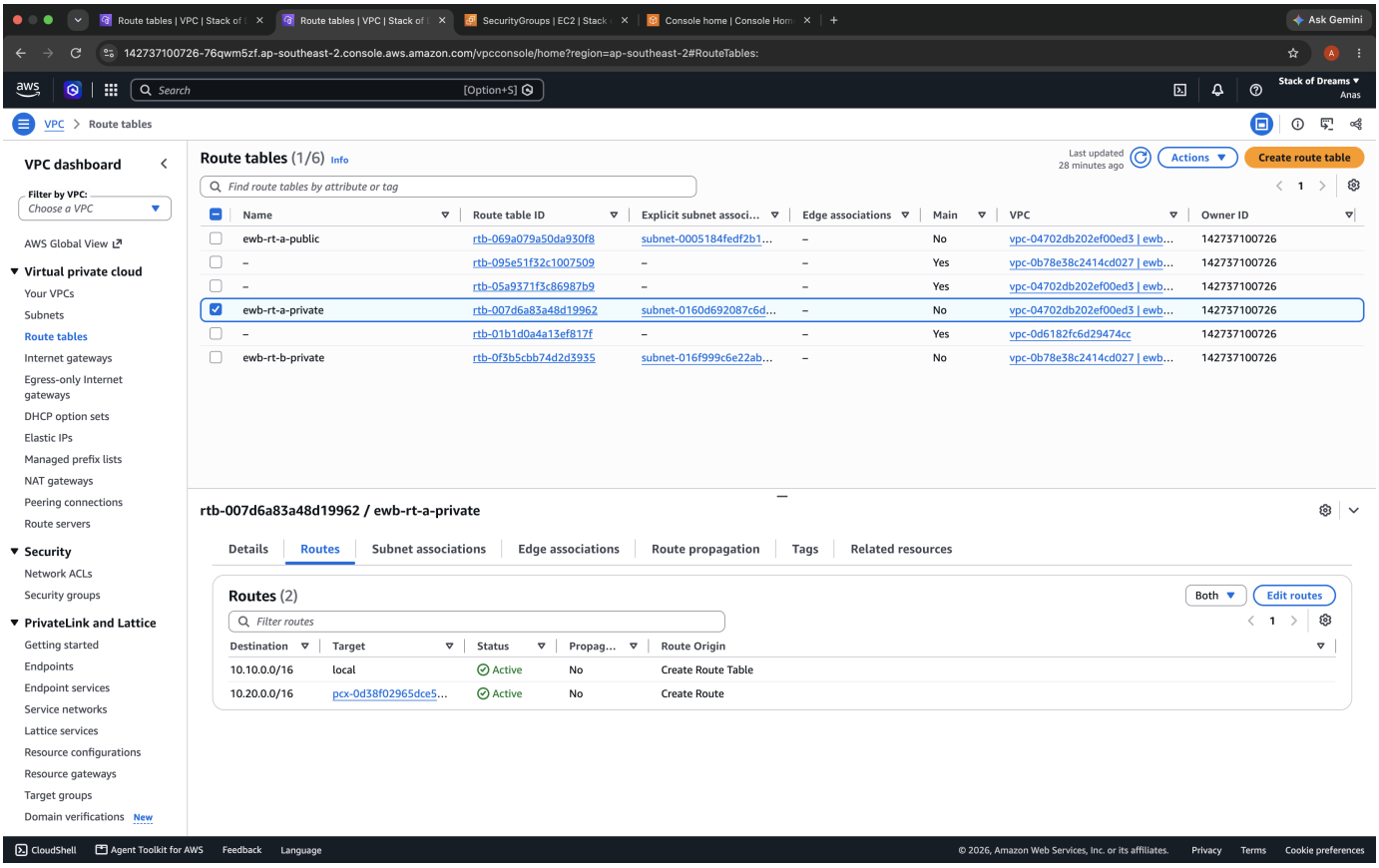
Evidence screenshot supplied for the VPC peering connection status.




File: 05-Active-VPC-Peering.png

# 6. VPC A private route table

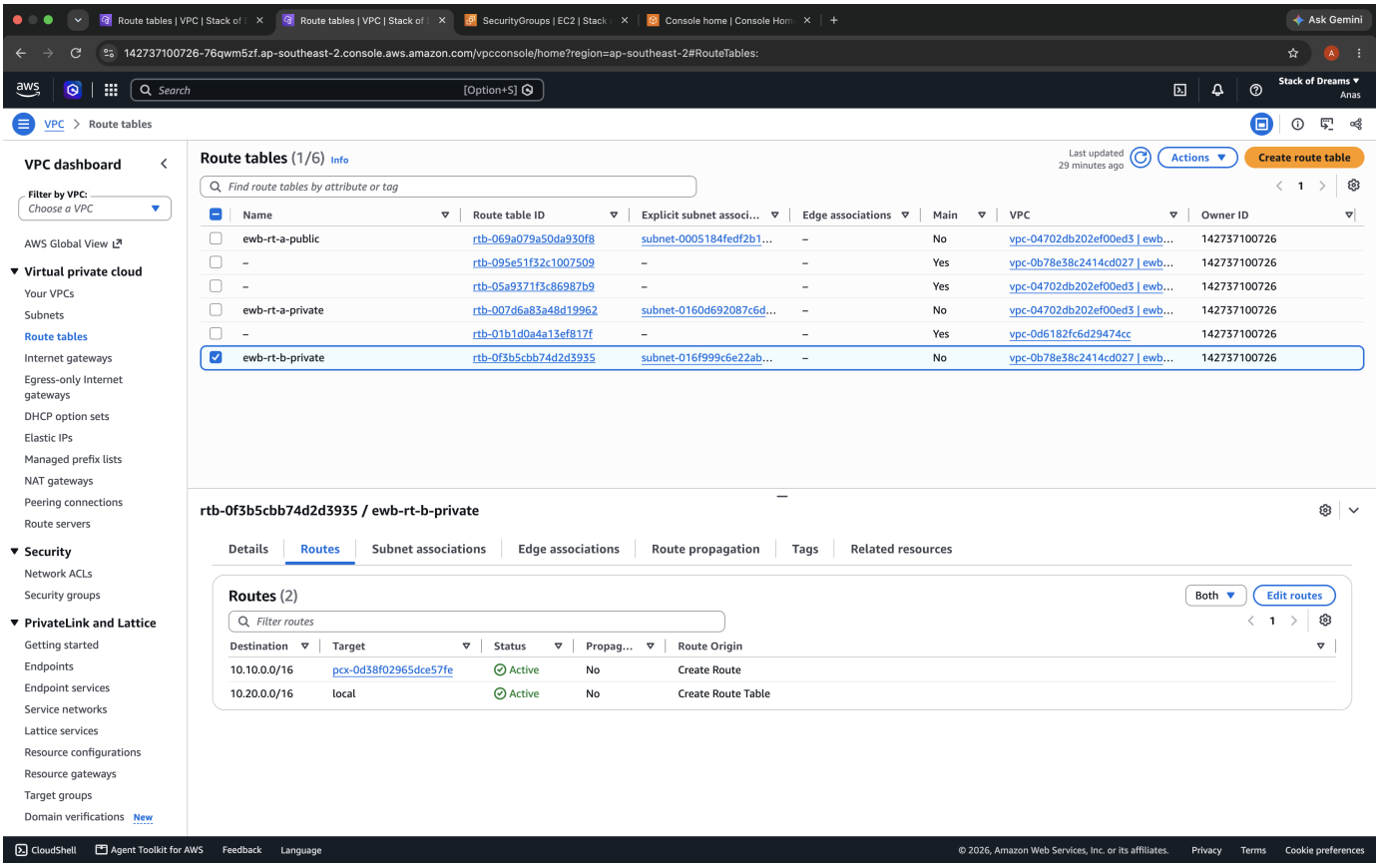
Route-table evidence for ewb-rt-a-private.




File: 06-ewb-rt-a-private.png

# 7. VPC B private route table

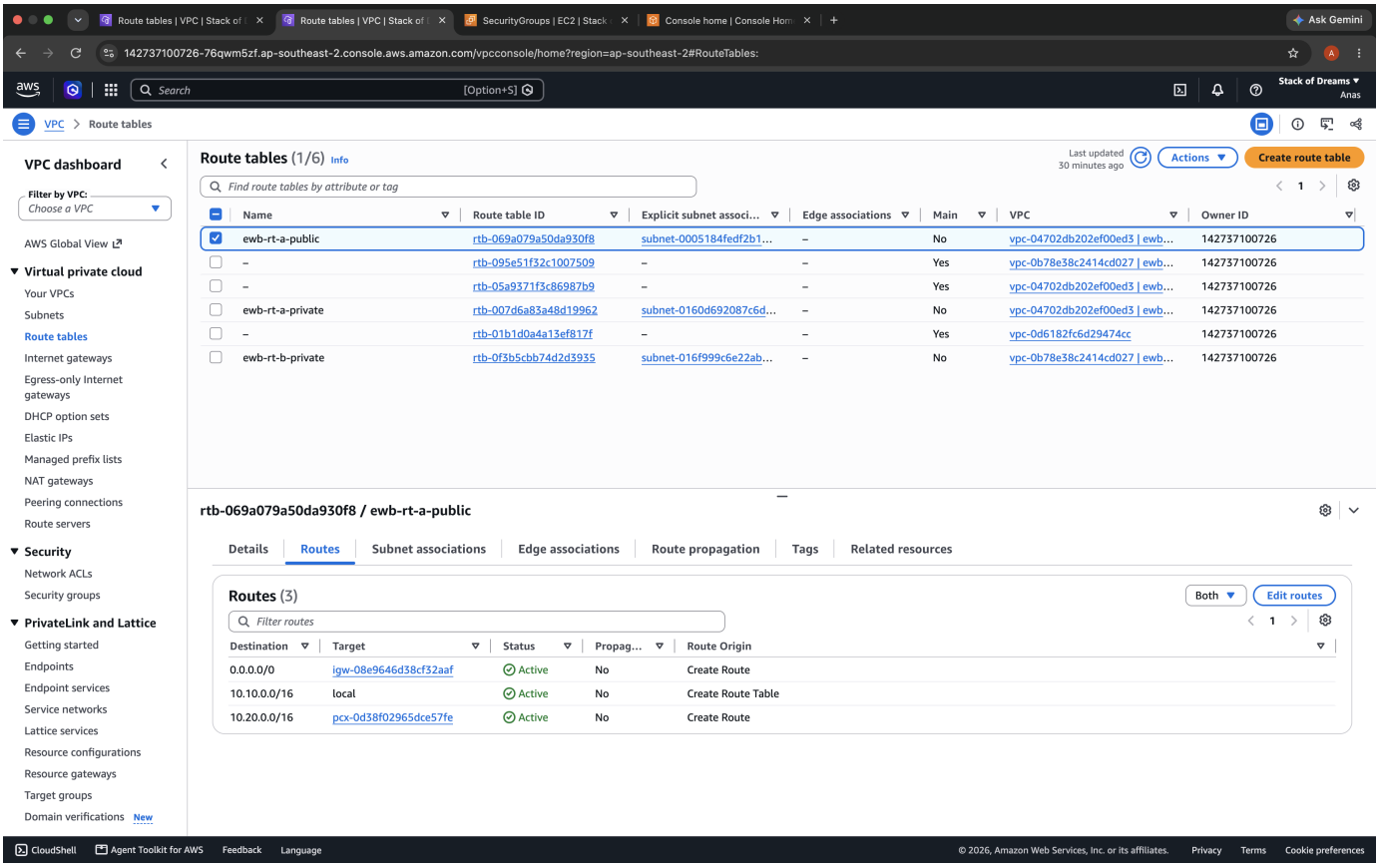
Route-table evidence for ewb-rt-b-private.




File: 07-ewb-rt-b-private.png

# 8. VPC A public route table

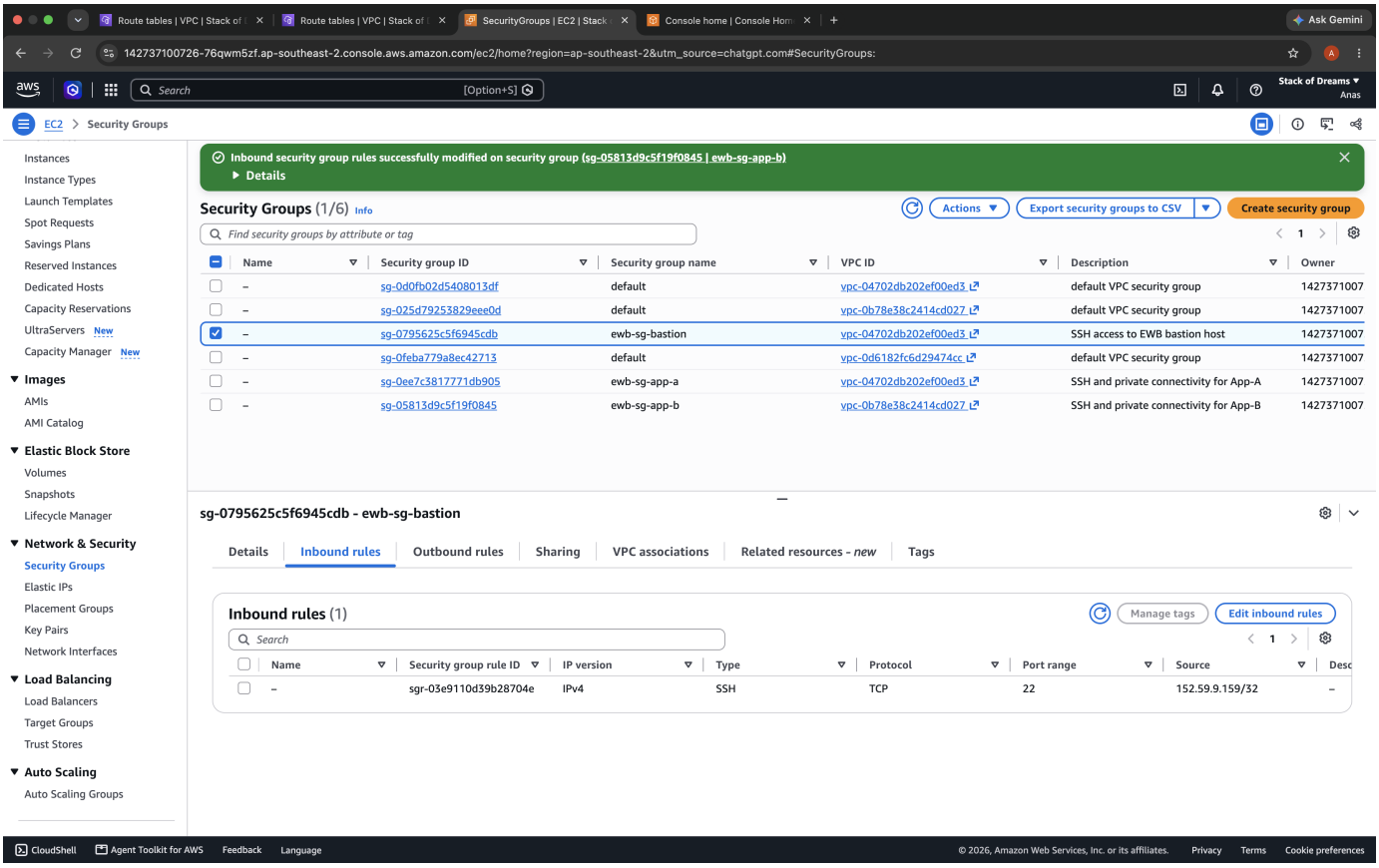
Route-table evidence for ewb-rt-a-public, including Internet Gateway and peering routes.




File: 08-ewb-rt-a-public.png

# 9. Bastion security group

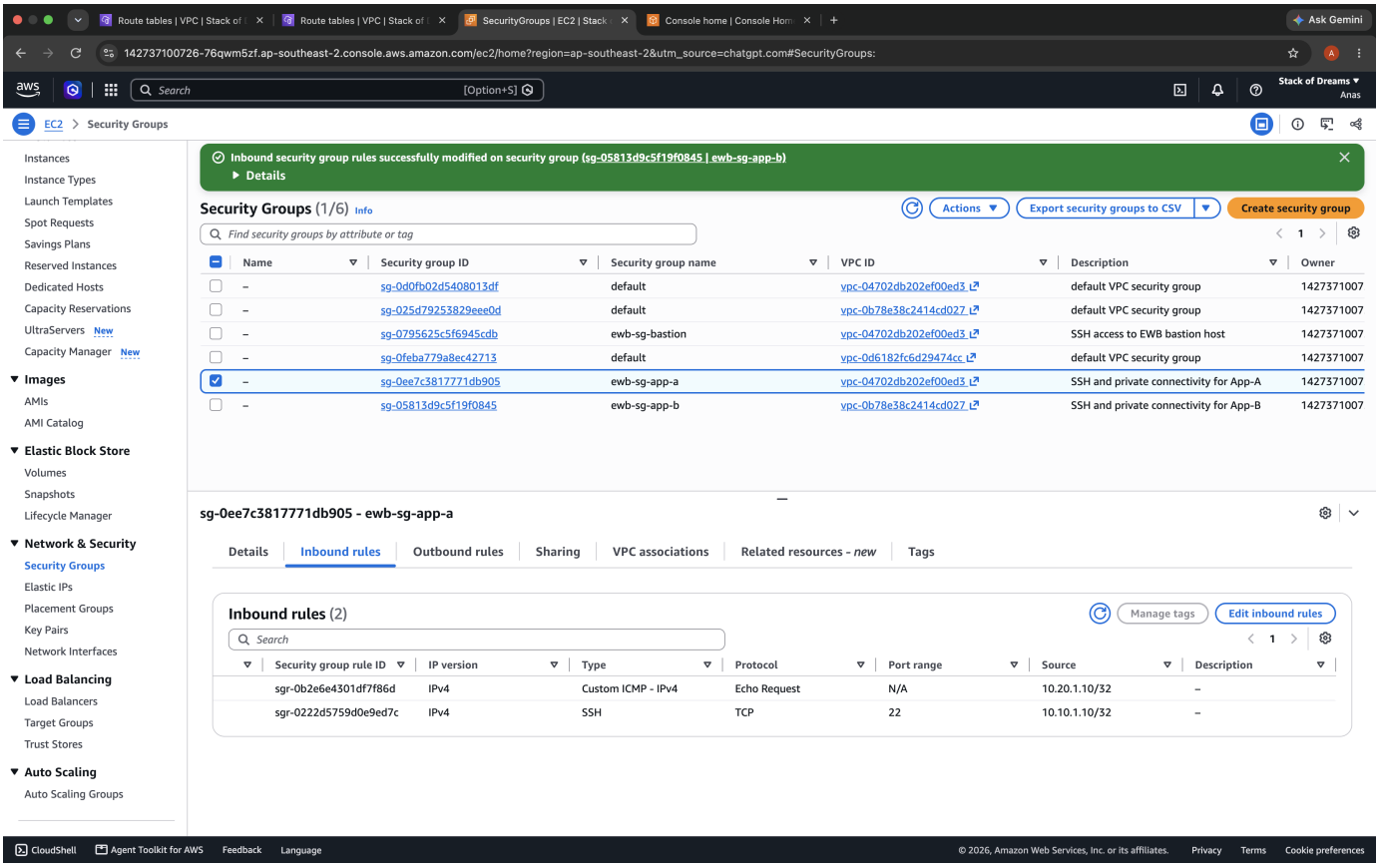
Inbound-rule evidence for ewb-sg-bastion.




File: 09a-ewb-sg-bastion.png

# 10. App-A security group

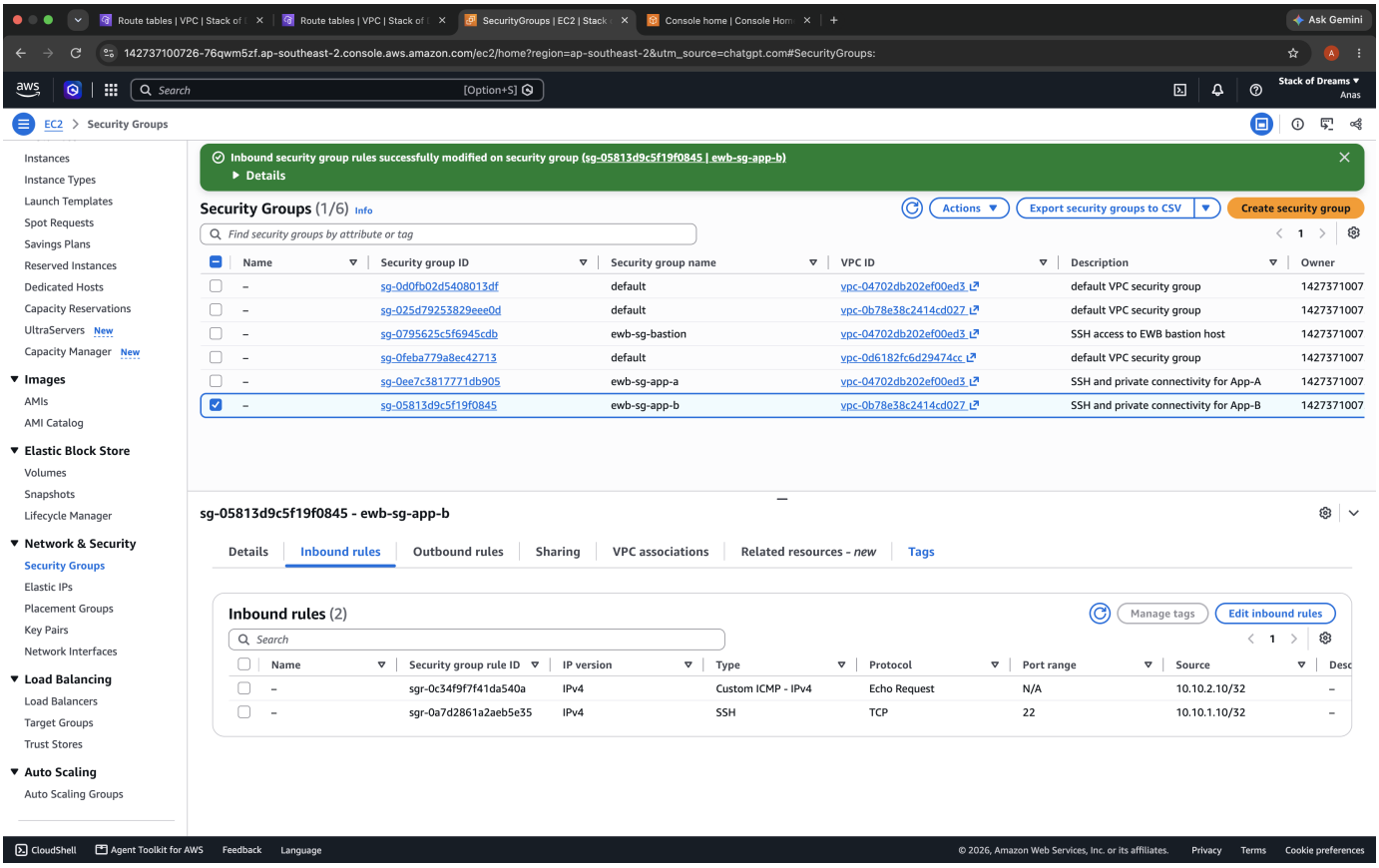
Inbound-rule evidence for ewb-sg-app-a.




File: 09b-ewb-sg-app-a.png

# 11. App-B security group

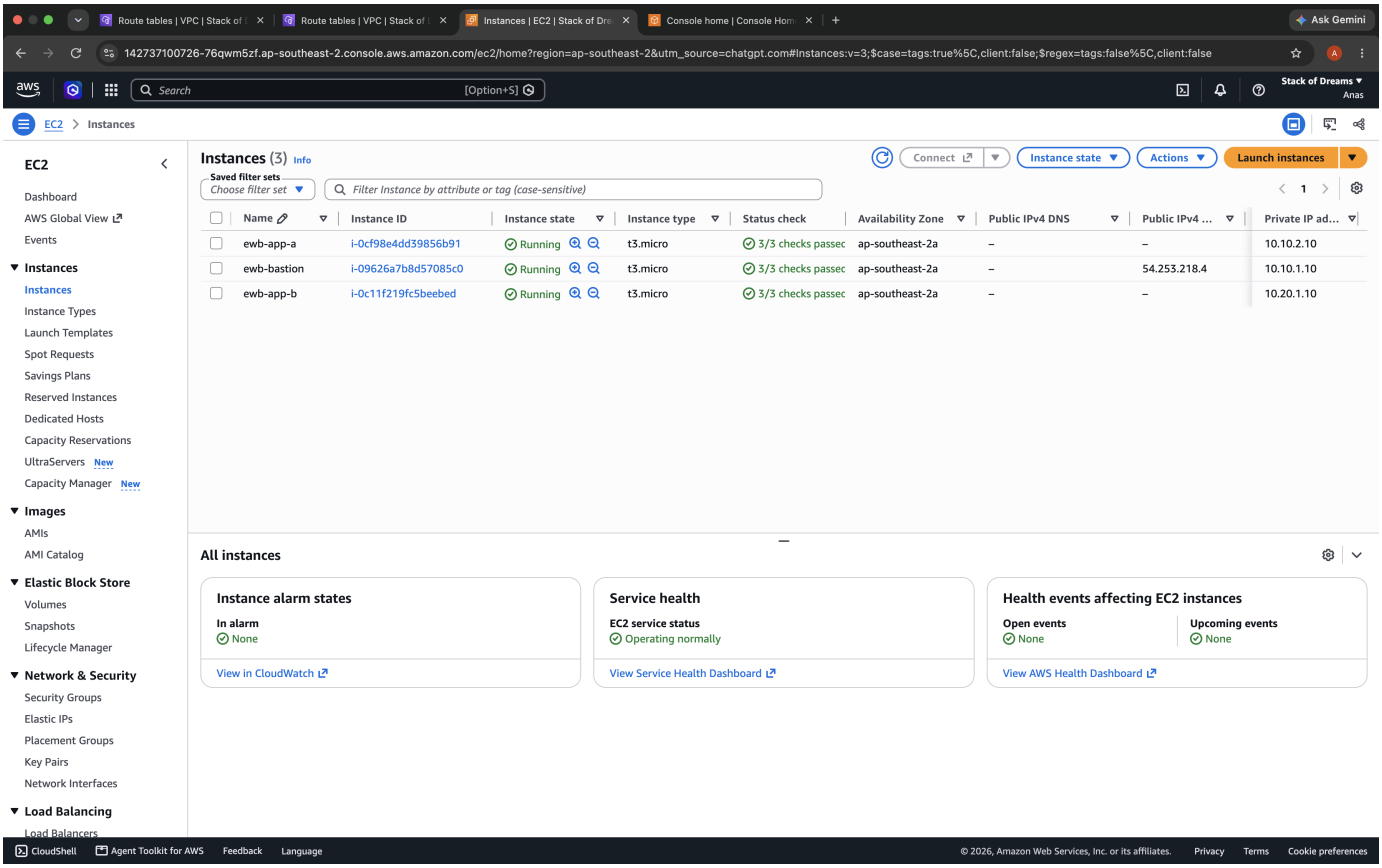
Inbound-rule evidence for ewb-sg-app-b.




File: 09c-ewb-sg-app-b.png

# 12. EC2 instance address verification

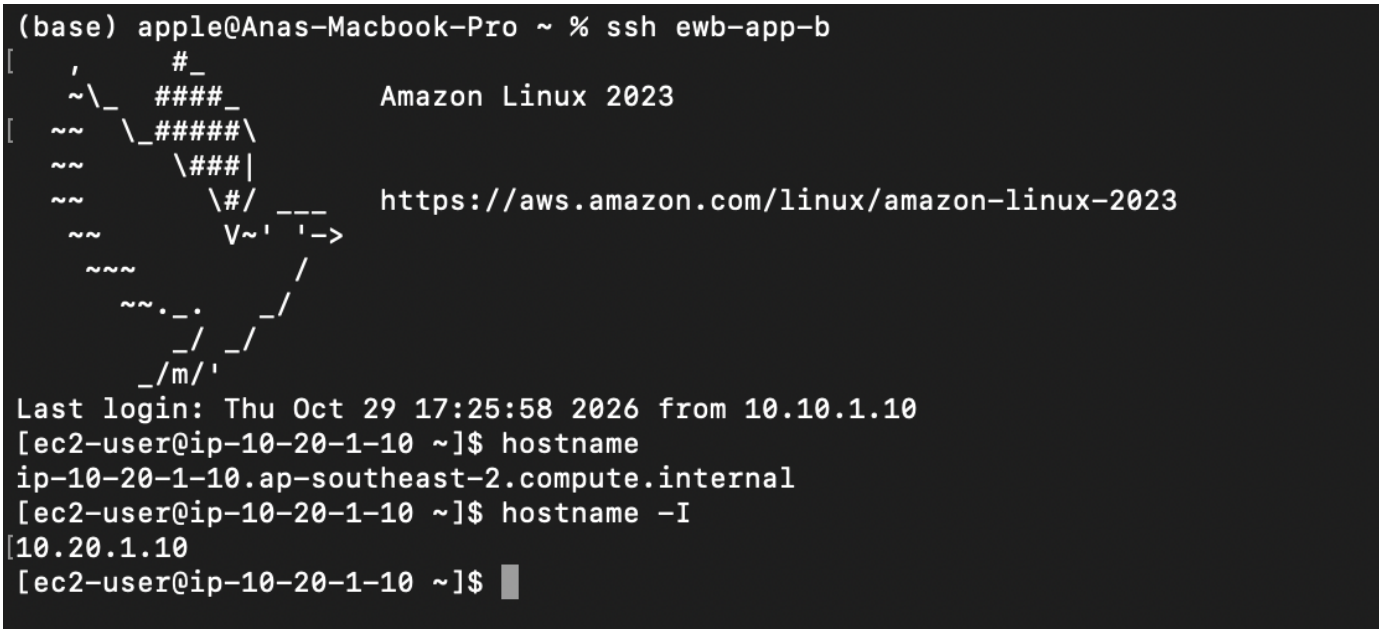
Evidence screenshot of the three EC2 instances and public-IP visibility. The expected private IPs are listed in the inventory below.




File: 10-EC2-Private-IP-Verification.png

## 13. SSH access to App-B through Bastion

Terminal evidence supplied for SSH access to the private App-B instance through the Bastion host.




File: 11-SSH-App-B-via-Bastion.png